

LAKSHITA SINGH



SOFTWARE ENGINEER | DISTRIBUTED SYSTEMS | ROBOTICS & AI

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PROFESSIONAL SUMMARY

Software Engineer with a focus on scalable infrastructure, robotics, and AI. Experienced in designing production-grade control planes and high-availability services within large-scale distributed environments.

EXPERIENCE

Microsoft AI, Redmond | Software Engineer

June 2025 - Present

- Built and operated **distributed backend infrastructure** supporting **500+ internal developers**, owning the full **software development lifecycle (SDLC)** including deployment, routing, reliability, and production operations across 5 environments.
- Re-architected **operational workflows** by standardizing diagnostics; reduced developer onboarding latency by **75%**.
- Engineered **concurrency-safe reliability controls**, eliminating recurring failures and mitigating **\$200K+ in LiveSite impact**.
- Led a full-scale San Jose **data-center routing migration** and modernized **30+ libraries** (Core XT to .NET 8).
- Served as **Production DRI** for 20+ major incidents (latency, traffic degradation, pipeline failures); reduced resolution time by **83%** via **telemetry-driven root-cause analysis** using **Kusto (KQL)** and **Azure** monitoring.
- Contributed to ongoing R&D of a **patent-pending "Portless Connection"** approach for secure MCP-based prod workflows, supporting Microsoft AI Ads (**\$13B+ revenue**) and built **5+ AI-powered workflows** recognized as **"Top 3"** org-wide.

Microsoft, Redmond | Robotics Software Researcher

Sep 2024 – March 2025

Publication in Progress, Masters Program 6-Month Capstone Project with Microsoft Research Team

- Architected e2e robotic-arm manipulation with computer vision, **~86% accuracy** in object detection and pose estimation.
- Developed benchmarking framework improving model selection, evaluating SLMs for enhanced robotic decision-making.

Alzi , Alzheimer's companion app | CEO & Cofounder

Jan 2024 – March 2025

Finalist Dempsey Startup Competition'24, Buerk Center for Entrepreneurship, UW Foster School of Business.

- Targeting **1.3M individuals** in the US with early-stage Alzheimer's and their caregivers. Led the end-to-end development of GenAI-powered chatbot, leveraging OpenAI, Django, and React, to monitor behavior and mental health.

Personal Robotics Lab , UW | Student Researcher

Jan 2024 – June 2025

- Developed intelligent robotic feeding system with haptic feedback & improved adaptability for food preferences by **25%**, utilizing reinforcement learning algorithms (ϵ -greedy, LinUCB) for optimized strategy development.

Google (ExploreCS Program) | Research Intern

May 2022 - Jul 2022

- Developed **MLOps-based** testing pipeline, integrated **driving safety algorithms** for self-driving project, streamlining analysis.

PROJECTS

- Proof of Robotic Work (PoRW)** : **Top 10 team at Base Batches'26 Robotics Track**, Co-built privacy-preserving audit infra for autonomous robot fleets, compressing **37 MB of raw footage into 3.6 KB ZK proofs**; designed the full stack (robot → edge inference → cryptographic proof → blockchain) to verify actions without exposing raw sensor data.
- Synergizing Reinforcement Learning for Cognitive Medical Decision-Making** : Investigated continuous state-space action models for efficient Sepsis treatment using Dueling Double-Deep Q networks, resulting in a 15% improvement in predicted outcomes. Awarded Best Paper at IVUS'23  and published as First Author in CCIS, Springer.

SKILLS

LANGUAGES: C++/C, .NET, Python, JavaScript/Typescript, SQL, Kusto (KQL), YAML

TOOLS & FRAMEWORKS: React, Django, Node.js, MongoDB, Azure, PowerShell, Linux, Git, ROS, Docker

RELEVANT ENGINEERING AREAS: Distributed Systems, Production Control Planes, High-Availability Services, Large-Scale Cloud Infrastructure, MCP-Based Tooling Design, Automation of Operational Tasks, Research-Driven Prototyping

EDUCATION

University of Washington (UW) | M.S. Technology Innovation (CS Track - Robotics) | 2025

Intensive technical program focused on software architecture, robotics engineering, and full-stack system design.

Delhi Technological University (DTU) | B.Tech in Mathematics and Computing | 2023

ACHIEVEMENTS

- Competitions & Research:** Winner of 8+ hackathons like (WEBS'25, DeveloperWeek'24, Codefest NY) • Top scorer, Bitgrit Tweets Prediction Challenge • Selected in ACM NLP Summer School (Top 50 nationwide)
- Leadership:** Technology Ambassador Fellow'23-25, GIX-UW • Harvard WECODE'22 Scholar • Vice-President, MACS (DTU)
- Extracurriculars:** Semi-Professional Actor (30+ stage plays) • Poetry & Open Mics (Local Bellevue area)